When Trust Doesn't Transfer: Diagnosing Source-Credibility Shortcuts in Fact Verification Models

Anonymous ACL submission

Abstract

Fact-verification models weighted by source trustworthiness need evaluations that only a trust-using model can pass. We audit a representative source-trust system and find three failures: a synthetic split leaks the verdict through textual register (trust-blind accuracy 0.996), AVeriTeC accuracy equals the majority-class prior (macro-F1 0.265), and a simulation split is closed-world trivial. We construct a synthetic, controlled trust-isolating benchmark holding evidence text byte-identical across trust tiers with source category fixed. The result is a sharp inversion: the trust-blind baseline collapses to chance (macro-F1 0.385 ± 0.007), while a trust-aware graph network wins (v2-HGNN 0.918 ± 0.017) and a logistic-regression probe with the trust scalar reaches 0.882. An NLI-enhanced variant (v3-NLI, 0.907 ± 0.019) scores below v2-HGNN because this benchmark fixes inference strength at 1.0, removing the axis NLI would otherwise improve. Mid-tier boundary probes (ST $\in \{0.50, 0.62, 0.72\}$) further reveal that even a trust-aware probe collapses to macro-F1 0.204, confirming that scalar access alone is insufficient and trust must be reasoned over continuously. We release the benchmark and generator with this submission.

1 Introduction

Automated fact verification predicts whether a claim is supported, refuted, or unverifiable given retrieved evidence. A persistent failure mode is shortcut learning: models reach high accuracy by exploiting surface regularities rather than reasoning over evidence. The best known case is claim-only bias in FEVER, where a classifier that never reads the evidence nearly matches an evidence-aware model (Schuster et al., 2019). A natural and well-motivated direction is to make verification *source-aware*: identical text from a peer-reviewed article and from an anonymous post should not be treated identically, so evidence should be weighted by the trustworthiness of its source (Popat et al., 2018; Baly et al., 2018).

This paper is about how that idea is *measured*. To claim that a model uses source trust, one needs an evaluation that only a trust-using model can solve. The intuitive construction pairs the same claim with a high-trust and a low-trust source and labels the low-trust version not-enough-evidence. We show that a careful instance of this construction silently leaks the verdict, that the leakage makes a trust-blind model appear to succeed, and that a second common reporting choice, accuracy on a class-imbalanced benchmark, hides a majority-class predictor. We then show how to build an evaluation that does isolate source trust, and what changes when it is used.

Our study object is a representative source-trust FV system that computes a per-evidence epistemic confidence from source trust, an evidence-type weight, and an inference strength, and aggregates these into a verdict through a heterogeneous graph neural network with a symbolic and a neural decision layer. We audit its evaluation, find it does not measure what it claims, and rebuild the measurement. Figure 3 summarizes the corrected benchmark.

Contributions.

1. **A leakage diagnosis.** We show that a synthetic "shortcut-breaking" trust split leaks the verdict through textual register (a template field predicts it at 0.99 and a trust-blind model scores 0.996), that reporting AVeriTeC accuracy of 0.665 masks the majority-class prior (macro-F1 0.265), and that the simulation split is closed-world trivial (Section 4).

2. **A trust-isolating benchmark.** We construct a split that admits no detectable non-trust path to the label, and report the lesson that text isolation alone is insufficient: a first version still

leaked at macro-F1 0.82 through the source-category proxy until the category was neutralized (Section 5). The benchmark is deliberately single-evidence, single evidence-type (testimony), and single source-category; these constraints hold every axis except source trust constant so that results are attributable to trust alone.

3. **The inversion.** On the corrected benchmark a trust-blind model is at chance and a trust-aware model wins, the opposite of the leaky split, confirmed with both a controlled probe and the original graph network (Section 6).

We release the benchmark generator and data so the construction can be inspected and extended.

2 Background and Related Work

Fact-verification benchmarks. FEVER introduced large-scale claim verification over Wikipedia with three verdict classes (Thorne et al., 2018). AVeriTeC moved to real-world claims with web evidence and four verdict classes, and is heavily skewed toward the refuted class (Schlichtkrull et al., 2023). AVeriTeC distinguishes plain verdict accuracy from an evidence-gated score; we use it only as a verdict-accuracy testbed and report macro-F1 against the majority baseline, because the class skew makes accuracy alone uninformative.

Shortcuts and robustness in FV. Claim-only bias was documented for FEVER together with a symmetric evaluation set (Schuster et al., 2019). Annotation artifacts — surface patterns in hypotheses that predict labels without evidence — are a broader class of the same problem (Gururangan et al., 2018), and hypothesis-only baselines have exposed such artifacts in NLI benchmarks (Poliak et al., 2018). VitaminC built contrastive evidence pairs that force a model to read the evidence rather than the claim (Schuster et al., 2021). These works target lexical and evidence-content shortcuts. The shortcut we study is different: a model can recover the verdict from the *register* of the evidence text, which is correlated with source trust by construction. Our corrected benchmark instantiates the contrast-set principle (Gardner et al., 2020) — a paired set of records differing in exactly one signal, so that a model can pass only by reading that signal — specialised to the source-trust axis.

Source-credibility verification. DeClarE weighted evidence with a learned source-trustworthiness representation for claim credibility (Popat et al., 2018). Source factuality has been predicted at the outlet level from multiple signals (Baly et al., 2018). These supply or use a trust signal but do not provide a measurement that isolates it; our contribution is the isolating evaluation.

Graph and symbolic FV. Evidence aggregation over graphs is a standard FV backbone, from GEAR (Zhou et al., 2019) and the kernel graph attention network (Liu et al., 2020) to stance-aware aggregation (Si et al., 2021) and heterogeneous-graph reasoning over text and tables (Gong et al., 2024). Knowledge-graph-based grounding checks have been applied to detect hallucinations in LLM outputs (Sansford et al., 2024), and logic-regularized models give interpretable, auditable verdicts (Chen et al., 2022). The system we audit sits in this family; our focus is its evaluation, not its architecture. Epistemology-grounded reasoning in language models is nascent and, to date, targets general deduction rather than fact verification (Sathish, 2026).

3 The System Under Study

The audited system is a prior implementation (Anonymous, 2026a); its leaky-split, AVeriTeC, and simulation numbers (the Original split row of Table 4 and the upper rows of Table 1) are taken from that implementation's evaluation logs, with the corrected benchmark released with this submission. We treat the system as the study object rather than patching it in place: the benchmark contribution is separable from, and portable beyond, the specific architecture that prompted it. The system assigns each evidence item an epistemic confidence

$$\mathrm{EC}_i = 1 - (1 - \mathrm{ST}_i)^{\mathrm{EW}_i \cdot \mathrm{IS}_i}, \quad (1)$$

where $\mathrm{ST}_i \in [0, 1]$ is a registry-assigned source-trust score, $\mathrm{EW}_i \in [0, 1]$ is an evidence-type weight (testimony: 0.80; perception: 0.95; inference by analogy: 0.55), and $\mathrm{IS}_i \in [0, 1]$ is an inference strength. The single-evidence verdict rule is: if $\mathrm{EC}_i \geq 0.75$ and the evidence supports the claim, output SUPPORTED; if $\mathrm{EC}_i \geq 0.75$ and the evidence refutes it, output REFUTED; otherwise output NOT_ENOUGH_EVIDENCE (this rule defines the benchmark data labels; the trained model's inference path is described in Section 6). This threshold grounds the three tiers of the corrected benchmark: high-trust sources ($\mathrm{ST} \geq 0.85$, $\mathrm{EW} = 0.80$, $\mathrm{IS} = 1.0$) yield $\mathrm{EC} \geq 0.781 > 0.75$ and receive

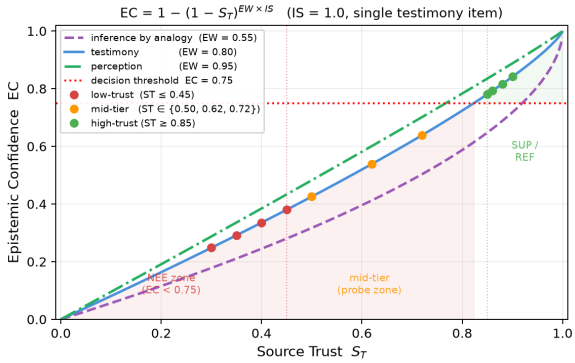

Figure 1: EC as a function of source trust S_T for the three evidence-type weights. The red dotted line marks the decision threshold (EC = 0.75). Coloured dots show the benchmark tier values. High-trust sources ($S_T \geq 0.85$, testimony curve) clear the threshold; low-trust ($S_T \leq 0.45$) fall well below it; mid-tier probes ($S_T \in \{0.50, 0.62, 0.72\}$) sit in the gap between the two zones, all still below 0.75.

a definitive verdict; low-trust sources (ST ≤ 0.45) yield EC $\leq 0.380 < 0.75$ and receive NEE; mid-tier sources (ST $\in \{0.50, 0.62, 0.72\}$) yield EC $\in \{0.426, 0.539, 0.639\}$, all below 0.75, and are used as boundary probes to test whether a model reasons continuously rather than learning a binary threshold.

Per-claim evidence is encoded by a heterogeneous graph network (Schlichtkrull et al., 2018) (graph attention (Veličković et al., 2018) over claim and evidence nodes) with stance and inference-strength heads, and the verdict is produced either by a symbolic threshold on the aggregated EC or by a learned verdict head. For this study the relevant property of Eq. (1) is monotonicity in ST: with evidence type and inference strength fixed, raising source trust raises confidence. A model that ignores ST has no access to this signal. Figure 1 visualises the formula.

4 Diagnosis: The Original Evaluation Does Not Measure Source Trust

We evaluate three claims in the original setup and find each undermined by its own data. Table 1 collects the numbers.

The synthetic split leaks through text register. The synthetic "shortcut-breaking" split pairs claims with high- and low-trust sources and labels low-trust supporting text as not-enough-evidence, so that, in principle, only source trust can separate the pair. In practice the generator draws low-trust evidence from a rumour register (for example, "An anonymous post claimed that…") and high-trust evidence from an official register ("published findings confirming…"). Because the register is determined by the trust tier and the trust tier determines the verdict, the text register determines the verdict. A single-feature lookup confirms this: a per-record template field predicts the verdict at 0.99 (the field value encodes verdict, trust tier, and source identity simultaneously — e.g., `high_trust_supported`, `low_trust_nee` — so 13 of 14 template types are 100% predictive), the source identifier at 0.82, and the raw evidence text and its stance at 0.92 and 0.54 respectively (Table 1, Figure 4). Consequently, a trust-blind model that reads only the text reaches 0.996 accuracy on this split, and the trust-aware model scores lower (0.889), because the leaky split rewards reading the register rather than reasoning about the source.

AVeriTeC accuracy hides the class prior. The AVeriTeC subset is heavily imbalanced. The system maps AVeriTeC's four verdict classes to three (supported, refuted, not-enough-evidence). On this development set, always predicting the majority class (refuted) yields accuracy 0.660 and macro-F1 0.265. A standard text classifier collapses onto the majority class, predicting the not-enough-evidence class exactly once out of 462 examples, for accuracy 0.656 and macro-F1 0.354. The reported system accuracy of 0.665 lies on this prior; without macro-F1 it cannot be distinguished from a majority predictor.

The simulation split is closed-world trivial. The simulation split is drawn from AI2-THOR (Kolve et al., 2017), an interactive 3-D environment that provides closed-world ground truth for object and state claims. On this split, every evidence item has the same maximal source trust and evidence-type weight, so Eq. (1) carries no discriminative signal, and the verdict is recoverable by matching a numeric value against simulator ground truth. Accuracy is 1.000, which reflects task triviality rather than epistemic reasoning, and inflates any average that includes it.

5 A Trust-Isolating Benchmark

We rebuild the evaluation so that source trust is the only feature that determines the label. Figure 3 shows the construction.

Trust-isolation criterion. A benchmark is *trust-isolating* if (i) every non-trust input feature takes

Table 1: Diagnosis of the original evaluation. Each row reports a number that undercuts a source-trust claim. Lookup values are the fraction of verdicts recovered from a single non-trust feature.

Finding	Measure	Value
Synthetic: template field	verdict lookup	0.99
Synthetic: source id	verdict lookup	0.82
Synthetic: trust-blind model	accuracy ↑	0.996
AVeriTeC: majority baseline	accuracy / macro-F1	0.660 / 0.265
AVeriTeC: text classifier	accuracy / macro-F1	0.656 / 0.354
Simulation split	accuracy ↑	1.000

the same value for both records in each source-trust pair, and (ii) no single non-trust feature predicts the verdict above chance on the test set. The corrected benchmark satisfies both conditions (Table 3); we verify this empirically after construction.

Design. For each claim we generate one neutral evidence sentence and attach it, byte-identical, to a high-trust and a low-trust source. The verdict is derived from the system's own confidence rule, Eq. (1): with a single testimony item, high trust ($\mathrm{ST} \geq 0.85$) yields supported or refuted, and low trust ($\mathrm{ST} \leq 0.45$) yields not-enough-evidence. The text is therefore constant within each high/low pair, and only the source-trust scalar flips the label. The benchmark has 2200 records, 1600 for training and 600 held out for test (480 standard pairs plus 120 mid-tier boundary probes); Figure 2 shows the three ST zones. Table 2 shows a concrete paired record.

Table 2: Paired record from the benchmark. *Claim*: "The Velora battery cell output is 318 megawatts." *Evidence*: "Recorded measurements state that the Velora battery cell output is 318 megawatts." Evidence text is byte-identical in both rows; only ST changes, flipping the verdict.

Source	ST	EC	Verdict
nm_h85a (high-trust)	0.85	0.781	SUPPORTED
nm_l30a (low-trust)	0.30	0.248	NEE

Text isolation is necessary but not sufficient. A first version of the benchmark held the text identical across tiers but allowed sources of different categories. It looked clean under single-feature checks, yet a trust-blind model still reached macro-F1 0.82 on the held-out test. The reason is an interaction the single-feature checks missed: the model reads the 6-dimensional source-category feature, the category is correlated with trust, and so it transfers to most held-out sources; combined with a "supporting text implies supported" default, it recovers the label without using the trust scalar. We therefore hold the source *category* constant (a single category for all records) and vary only the continuous trust scalar, with held-out trust values at test so that a trust-using model must treat trust as a continuous signal rather than memorize values. This is itself a lesson: isolating a feature requires controlling its proxies, not only the feature.

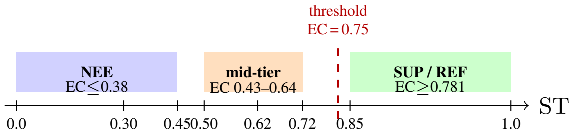

Figure 2: Trust tiers on the benchmark. Evidence text is byte-identical across all tiers; only the ST scalar changes. The red dashed line marks the $\mathrm{EC} = 0.75$ decision threshold; all high-trust records ($\mathrm{ST} \geq 0.85$, $\mathrm{EC} \geq 0.781$) clear it. Mid-tier probes ($\mathrm{EC} \in \{0.43\text{–}0.64\}$) fall below it, testing whether a model reasons continuously rather than thresholds ST into two bins.

Verification. The corrected benchmark admits no non-trust path to the label. Every standard-tier evidence text appears with more than one gold label (1028 of 1068 distinct texts; the remaining 40 are mid-tier boundary probes that carry only NEE labels by design), and every non-trust feature predicts the verdict at chance (Table 3). Figure 4 contrasts the per-feature leakage of the original and corrected splits.

Table 3: Trust-isolation verification on the corrected benchmark. A non-leaking feature predicts the verdict at the chance rate (0.50 for the high/low decision). Source trust is the only predictive signal, and it is held out by value at test.

Feature	verdict predictable
evidence text (full)	0.50
evidence text (prefix)	0.50
stance	0.50
source category	0.50 (constant)
source-trust scalar	predictive (held out at test)

6 Experiments: The Inversion

We evaluate three models on the corrected benchmark across three independent training runs each, with the majority and always-NEE baselines, reporting macro-F1 as the primary metric because of the class balance by construction. The text embedding is all-MiniLM-L6-v2 (Reimers and Gurevych,

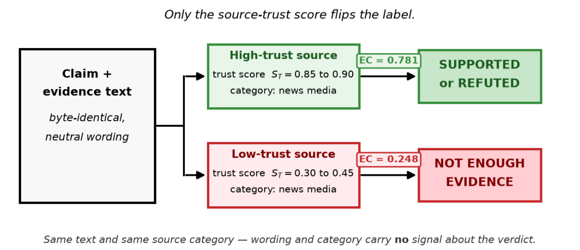

Figure 3: The trust-isolating benchmark. The same evidence text is attached to a high-trust and a low-trust source of the same category; only the source-trust scalar differs, so text and category carry no label signal and only a trust-using model can separate the pair.

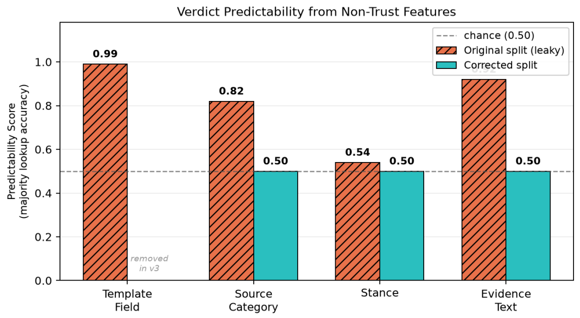

Figure 4: Verdict recoverable from a single non-trust feature. On the original split the template field predicts the verdict at 0.99 (the field value encodes verdict, trust tier, and source identity in a single string — 13 of 14 types are 100% predictive), source category at 0.82 (category correlated with trust tier), evidence text at 0.92 (text-register leak), and stance at 0.54. On the corrected split the template field is removed entirely; source category, stance, and evidence text all drop to chance (0.50), consistent with Table 3.

2019) (384 dimensions). All GNN variants share hidden dimension 256 and batch size 32, up to 30 epochs with early stopping (patience 5) on validation loss; per-model hyperparameters were selected by a 30-trial Optuna (Akiba et al., 2019) search (baseline: heads 4, dropout 0.2; v2-HGNN: heads 2, dropout 0.2; v3-NLI: heads 2, dropout 0.4). All reported metrics are computed from per-run result files released with this submission (Anonymous, 2026b).

Models. We evaluate three variants from the system under study. The **baseline** uses text embeddings and graph structure with no EC layer; it is trust-blind by construction. **v2-HGNN** adds the epistemic-confidence layer (Eq. (1)): when the aggregated EC confidence score exceeds the decision threshold (0.75), the verdict is derived symbolically (SUPPORTED or REFUTED); when both support and refute scores exceed 0.75 simultaneously, the *HybridVerdictHead* resolves the conflict; and when neither exceeds 0.75, the *HybridVerdictHead* provides the verdict. The HybridVerdictHead is a learned head trained on all three verdict classes — it is not an automatic NEE fallback. **v3-NLI** extends v2-HGNN by adding three-dimensional DeBERTa-v3-small (He et al., 2021) NLI probabilities (entailment, neutral, contradiction) as auxiliary input features. Figure 5 shows the key structural difference.

Controlled probe. To isolate feature access from model capacity, we train a logistic-regression probe on the same features the trust-blind graph network sees (a sentence embedding of the evidence text and a source-category indicator), and compare it to the same probe with the source-trust scalar added. On the full test set (480 standard pairs plus 120 mid-tier boundary probes), the trust-blind probe reaches macro-F1 0.389, well below the trust-aware ceiling. The trust-aware probe reaches 0.882 overall — near-perfect on standard pairs (0.997) but collapsing to 0.204 on mid-tier records. This collapse shows that scalar access alone is insufficient: a linear model learns to threshold ST into high and low bins from training data, and this binary rule fails on the mid-tier boundary zone where ST is intermediate.

GNN models. We then train the system's own models across three independent runs each. The trust-blind baseline plateaus at verdict accuracy near 0.50 in training and reaches macro-F1 0.385 ± 0.007 on the held-out test. An always-NEE predictor achieves macro-F1 0.250, confirming that the baseline is above the trivial floor but far below a trust-using model. The trust-aware v2-HGNN reaches macro-F1 0.918 ± 0.017. The gap of 0.533 $(0.918 - 0.385)$ is produced solely by access to source trust. A decision-path breakdown confirms the two-layer structure: the EC symbolic path handles 40% of test decisions (all high-trust records) at 100% accuracy; the remaining 60% are deferred to the HybridVerdictHead, which achieves 0.870 accuracy on the 360 low- and mid-trust (NEE) records.

v3-NLI and the NLI shortcut. The v3-NLI variant, which augments v2-HGNN with DeBERTa-v3-small NLI probabilities, reaches macro-F1 0.907 ± 0.019 — lower than v2-HGNN and with slightly higher variance. This outcome is specific to the IS-constant setting of this benchmark: NLI probabilities are a natural estimator for inference strength IS (how strongly evidence text binds to the

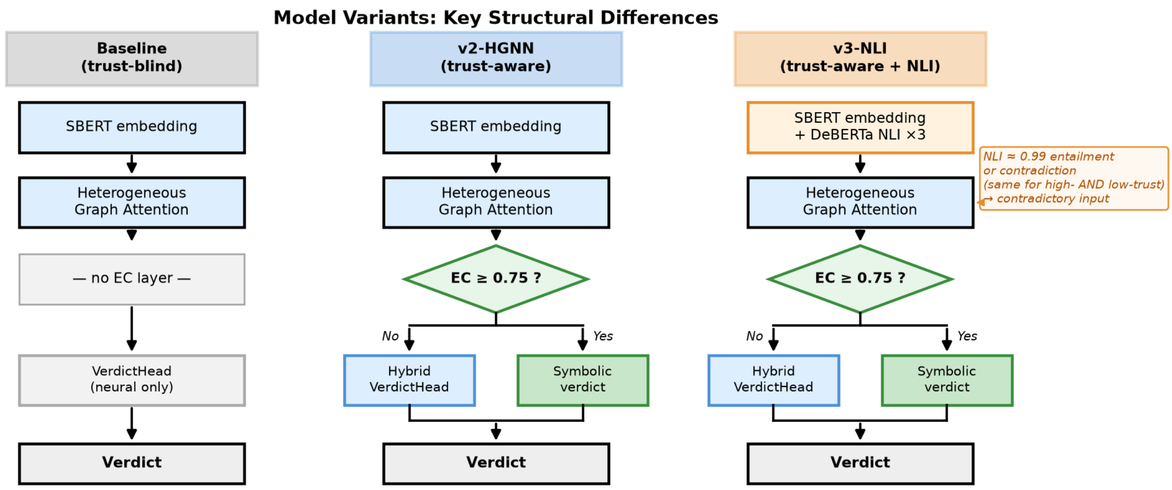

Figure 5: Key structural differences between the three model variants. Baseline has no EC layer and never reads source trust. v2-HGNN and v3-NLI both apply the EC decision (EC ≥ 0.75) symbolically; v3-NLI additionally conditions on DeBERTa NLI probabilities, which creates a competing text-semantic signal on benchmark template text.

claim), but the benchmark holds IS $= 1.0$ for all records, so NLI has no IS axis to contribute to. Instead, on this benchmark's template text, DeBERTa assigns near-perfect entailment or contradiction regardless of source trust, because all evidence text is semantically unambiguous by construction. This creates a contradictory training signal at the input level: the same near-perfect NLI features appear in both high-trust (SUPPORTED) and low-trust (NEE) records, because evidence text is byte-identical within each pair. The encoder receives identical NLI-dominated input for records with opposite labels, forcing it to learn to rely on the source-trust scalar instead — which competes with the strong text-semantic prior baked into the NLI columns. The decision-path breakdown localises the underperformance: the EC symbolic path is identical in both models (40% of decisions, 100% accuracy); the gap arises entirely in the HybridVerdictHead fallback, where v3-NLI achieves 0.852 accuracy on 360 NEE records compared to 0.870 for v2-HGNN, making on average 53 errors per run versus 47 — all SUPPORTED or REFUTED predictions on records that should be NEE, confirming that NLI features amplify the text-trust conflict specifically in the fallback path. NLI features would be expected to help in a setting where IS varies; the current benchmark holds IS constant, which is why they interfere rather than assist.

Mid-tier boundary probes. Records with ST $\in \{0.50, 0.62, 0.72\}$ (EC $\in$ $\{0.426, 0.539, 0.639\}$) fall below the 0.75 threshold and are labelled NEE. Per-record evaluation across the 600-record test set reveals that both v2-HGNN and v3-NLI achieve perfect macro-F1 (1.000) on the 480 standard records; *all errors come from mid-tier records*. v2-HGNN scores 0.252 ± 0.024 on mid-tier and v3-NLI scores 0.237 ± 0.026 — both above the trust-blind baseline (0.016) and above the LogReg trust-aware probe (0.204), but far below the standard-record ceiling. This shows that the mid-tier boundary zone is the *only* unsolved region: the EC layer handles high- and low-trust records reliably, but continuous reasoning at intermediate ST values remains an open challenge for all tested architectures.

The inversion. Table 4 and Figure 6 place the results beside the original leaky split. On the leaky split the trust-blind model wins (0.996 versus 0.889 accuracy), because the split rewards reading the text register. On the corrected split the trust-blind model collapses to chance and v2-HGNN wins. v3-NLI also outperforms the trust-blind baseline by a large margin but scores below v2-HGNN, showing that NLI-enriched models cannot bypass the trust-isolation constraint when text is informationally symmetric. The trust signal helps only once the evaluation stops leaking. This inversion is the central result: it shows both that the original evaluation measured the wrong thing and that, measured correctly, source trust does carry the claimed effect.

7 Conclusion

Source-trust reasoning in fact verification must be measured on evaluations that demonstrably require

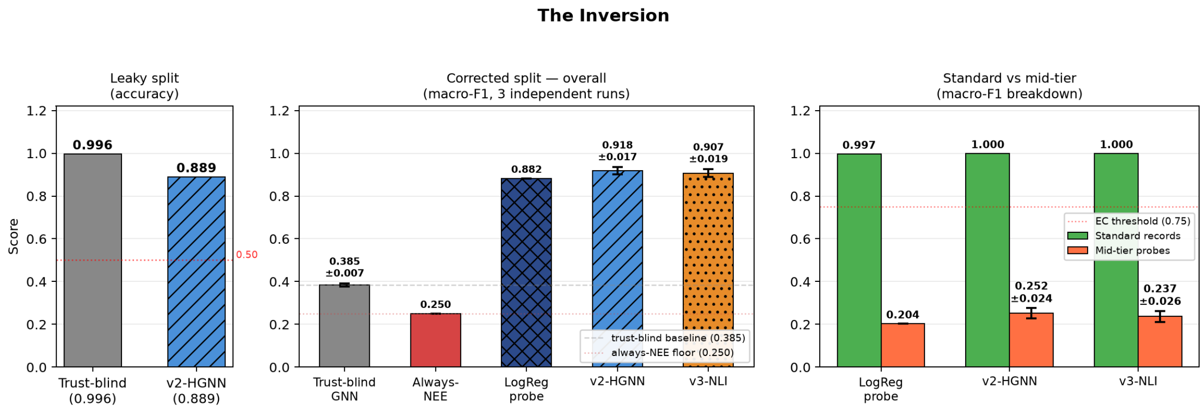

Figure 6: The inversion. On the original leaky split a trust-blind model wins on accuracy; on the corrected split it collapses to chance (macro-F1) while a trust-aware model wins, with both the controlled probe and the original graph network.

Table 4: Results on the trust-isolating benchmark. GNN results are mean ± std across three independent runs. Trust-blind wins only on the leaky split; on the corrected benchmark it collapses to chance. Standard records are perfectly classified by trust-aware GNNs; mid-tier boundary probes expose the remaining gap.

Setting	Metric	Trust-blind	Trust-aware	v3-NLI
Original split (leaky)	acc ↑	**0.996**	0.889	—
Always-NEE floor	macro-F1	0.250	—	—
Corrected, LogReg probe	macro-F1 ↑	0.389	**0.882**	—
Corrected, GNN (3 runs)	macro-F1 ↑	0.385±0.007	**0.918±0.017**	0.907±0.019
Standard only (GNN)	macro-F1 ↑	—	**1.000**	1.000
Mid-tier probes (LogReg)	macro-F1 ↑	0.016	**0.204**	—
Mid-tier probes (GNN)	macro-F1 ↑	—	0.252±0.024	0.237±0.026

source trust. We showed that a natural "shortcut-breaking" construction leaks the verdict through textual register, that accuracy on a skewed benchmark hides the majority-class prior, and that controlling a feature requires controlling its proxies as well. We release a synthetic, controlled trust-isolating benchmark and report an inversion: a trust-blind model wins on the leaky split and is at chance on the corrected one, while a trust-aware model wins only when the evaluation no longer leaks. These results hold under single-evidence, single evidence-type, single-category conditions; the same isolation method should be applied to the evidence-type and inference-strength axes before the broader epistemic confidence framing can be claimed.

8 Limitations

The corrected benchmark is a controlled instrument: single-evidence, single evidence-type (testimony), single source-category (news media), and built from synthetic fictional claims. All claims are synthetic and fictional (e.g., "The Velora battery cell output is 318 megawatts"); this is deliberate — fictional claims prevent factual contamination and allow byte-identical evidence pairing — but it means the benchmark measures trust sensitivity under controlled conditions, not open-web realism. Generalisation to multi-evidence, multi-category, real-claim settings is not tested here; that realism axis is better carried by benchmarks such as AVeriTeC, reported with macro-F1 and a majority baseline. The mid-tier boundary zone reveals a gap in continuous trust reasoning: no current architecture fully bridges the 0.204 mid-tier score, and purpose-built models for continuous epistemic reasoning are an open problem. Both EW and IS in Eq. (1) are held constant throughout; an evidence-type-isolating benchmark and an inference-strength-isolating benchmark are the natural next steps. The IS finding also suggests a rehabilitation path for v3-NLI: NLI probabilities are appropriate estimators for IS and may improve over v2-HGNN in a benchmark where IS genuinely varies.

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